

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1066

Repetto 138 kV
August 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1066. The Interconnection Customer proposes a 197.6 MW natural gas combined cycle facility interconnecting at the Repetto 138 kV substation in the Virginia area. The request was received on September 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 293 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,186,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: REPETTO SUBSTATION to ASH ST 138kV line
Contingency: MNTELLO to NEOSHO 138kV line
Loading: 169.1% (Rating: 218 MVA)
DFAX: 73.7%
Pre-project loading: 102.2%
2. Facility: ASH STREET to HANCOCK 138kV
Contingency: HONDO to MOONSHINE 138kV
Loading: 135.2% (Rating: 334 MVA)
DFAX: 75.5%
Pre-project loading: 90.6%
3. Facility: MONTELLO to NEOSHO 138kV
Contingency: REPETTO to ASH ST 138kV
Loading: 134.9% (Rating: 257 MVA)
DFAX: 61.5%
Pre-project loading: 87.7%
4. Facility: ASH ST to MONTELLO 138kV line
Contingency: ASH ST to REPETTO SUBSTATION 138kV
Loading: 134.9% (Rating: 257 MVA)
DFAX: 61.5%
Pre-project loading: 87.7%
5. Facility: INGERSOLL RAND to MOONSHINE 138kV L
Contingency: HANCOCK to ASH ST SUBSTATION 138kV
Loading: 123.6% (Rating: 221 MVA)
DFAX: 33.7%
Pre-project loading: 93.5%

6. Facility: COLLEGE to POPLAR-ANTELOPE 138kV
 Contingency: HANCOCK to ASH ST 138kV Line
 Loading: 122.5% (Rating: 160 MVA)
 DFAX: 24.4%
 Pre-project loading: 92.3%
7. Facility: HONDO to MOONSHINE 138kV
 Contingency: HANCOCK SUBSTATION to ASH ST 138kV
 Loading: 113.9% (Rating: 251 MVA)
 DFAX: 33.7%
 Pre-project loading: 87.3%
8. Facility: BEARDSLEY ST to POPLAR-ANTELOPE 138kV
 Contingency: HANCOCK to ASH STREET 138kV
 Loading: 106.8% (Rating: 211 MVA)
 DFAX: 24.4%
 Pre-project loading: 84.0%
9. Facility: HUNTINGTON COURT to INGERSOLL RAND SUBSTATION 138kV L
 Contingency: ASH ST to HANCOCK 138kV
 Loading: 101.1% (Rating: 240 MVA)
 DFAX: 33.7%
 Pre-project loading: 73.4%
10. Facility: LEWISTOWN NO. 1 to HONDO SUBSTATION 138kV
 Contingency: HANCOCK to ASH ST 138kV
 Loading: 100.9% (Rating: 260 MVA)
 DFAX: 47.2%
 Pre-project loading: 65.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ASH ST to REPETTO 138kV	Rebuild line with higher-capacity conductor	\$865,000	60% (\$522,000)
HNCOCK SUBSTATION to ASH ST 138kV	Rebuild line with higher-capacity conductor	\$3,731,000	50% (\$1,853,000)
MONTELLO to NEOSHO 138kV	Rebuild line with higher-capacity conductor	\$1,480,000	56% (\$827,000)
ASH ST to MONTELLO 138kV	Rebuild line with higher-capacity conductor	\$1,564,000	56% (\$874,000)
MOONSHINE to INGERSOLL RAND 138kV	Rebuild line with higher-capacity conductor	\$1,807,000	34% (\$623,000)
COLLEGE to POPLAR-ANTELOPE 138kV	Rebuild line with higher-capacity conductor	\$3,411,000	46% (\$1,568,000)
HONDO to MOONSHINE 138kV	Rebuild line with higher-capacity conductor	\$3,540,000	34% (\$1,221,000)
BEARDSLEY ST to POPLAR-ANTELOPE 138kV	Reconductor line	\$1,922,000	46% (\$884,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUNTINGTON COURT to INGERSOLL RAND 138kV L	Reconductor line	\$1,060,000	34% (\$366,000)
LEWISTOWN NO. 1 to HONDO 138kV	Reconductor line	\$1,209,000	37% (\$448,000)
	Total allocated to project		\$9,186,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.